

RAJIB KHAN

Artificial Intelligence & Natural Language Processing Researcher | Dhaka, Bangladesh
rajibkhan4438@gmail.com | +880 1519-600174 | linkedin.com/in/rajib-khan-b924a1420 | github.com/Rajib4438

RESEARCH PROFILE

- Computer Science and Engineering graduate with research experience in low-resource Bengali NLP, multi-label classification, sentiment analysis, interpretable machine learning, and dataset development.
- Author or co-author of nine journal and IEEE conference papers, including a Data in Brief dataset article.
- Combines research practice with full-stack software development experience and seeks advanced study in Artificial Intelligence, Data Science, and Natural Language Processing.

EDUCATION

International University of Business Agriculture and Technology (IUBAT), Dhaka, Bangladesh

Bachelor of Computer Science and Engineering (BCSE) | Jan 2022-Apr 2026

CGPA: 3.47/4.00 | Major CGPA: 3.53/4.00

- Thesis: "Enhancing Multi-Label Classification in Bengali Contextual Questions using BERT-based Architecture."
- Relevant coursework: Data Structures and Algorithms, Database Management Systems, Software Engineering, System Analysis and Design, Operating Systems, Compiler Design, Management Information Systems, Artificial Intelligence and Machine Learning.

Higher Secondary Certificate (Science) | 2020

Secondary School Certificate | 2018

RESEARCH EXPERIENCE

Undergraduate Researcher - NLP and Machine Learning | IUBAT

2023-2026

- Research Advisor: Tanjim Taharat Aurpa.
- Developed a low-resource Bengali multi-label classification framework using XLM-RoBERTa and supervised contrastive learning.
- Curated PROSHNO BINNASH, a contextual multi-label question-answering dataset containing 4,069 entries derived from authentic Bengali texts; this work led to a Data in Brief publication.
- Collected and processed social-media data for sentiment-analysis studies on Dhaka Metro Rail, the Elevated Expressway, and the Padma Bridge.
- Constructed a Bengali IT-service review dataset and contributed to an interpretable aspect-based sentiment analysis study.
- Performed literature review, dataset development, preprocessing, model implementation, experimentation, model evaluation, result analysis, and academic writing.
- Presented research findings at IEEE QPAIN 2026 and IEEE SPICSCON 2025.

SOFTWARE ENGINEERING EXPERIENCE

Software Engineer Intern / Final-Year Practicum | TechStdio

Oct 2025-Jan 2026

- Designed and developed a multi-vendor e-commerce platform using ASP.NET Core Web API, SQL Server, and Angular 17.
- Built a seller dashboard for area-based rider assignment and decentralized order fulfillment.
- Integrated the bKash payment API, secure checkout logic, and automated PDF invoice generation.
- Strengthened practical skills in REST API integration, database management, teamwork, communication, and scalable software development.

SELECTED PUBLICATIONS

- Khan, R., Aurpa, T. T., Rahman, M. S., & Ashrafuzzaman, M. (2025). "PROSHNO BINNASH: Contextual Multi-Label Question Answering Dataset for Low-Resource NLP." *Data in Brief*.
- Khan, R., Rakib, A. B., Haque, O., Tabassum, M., Rahman, M. S., & Akter, M. (2026). "Aspect-Based Sentiment Analysis of Bangladeshi IT Service Reviews Using Transformer Architectures and Post-Hoc Interpretability." *IEEE QPAIN*.
- Islam, M. A., Khan, R., Rahman, M. S., Emu, S. A., & Anik, M. A. H. (2026). "BanglishBERT-CNN-MTL: A Hybrid Multi-Task Learning Framework for Aspect-Based Sentiment Analysis in Low-Resource Banglish Text." *IEEE QPAIN*.
- Tabassum, M., Haque, O., Khan, R., Rahman, M. S., & Akter, M. (2026). "Emotion-Aware Token Reweighting for Aspect-Based Emotion Detection in Bangla Text." *IEEE QPAIN*.
- Akter, M., Haque, O., Tabassum, M., & Khan, R. (2026). "A Hybrid Stacking-Based Framework for Multiclass Road Accident Severity Prediction with Interpretable SHAP Analysis." *IEEE QPAIN*.
- Haque, O., Hossain, A., Tabassum, M., Khan, R., & Tamim, F. S. (2025). "BanglaBERT-BiLSTM: A Novel Hybrid Architecture for Multiclass Mental Health Classification in Bengali Social Media Text." *IEEE ICCIT*.
- Aurpa, T. T., Jeba, S. M., Khan, R., Ahmed, M. S., & Dash, S. R. (2025). "Deep Learning Analysis of Public Sentiment on the Padma Bridge Through YouTube Comments." *IEEE SPICSCON*.
- Jeba, S. M., Khan, R., Ahmed, M. S., Aurpa, T. T., & Dash, S. R. (2024). "Sentiment Analysis on the Elevated Expressway in Bangladesh: A Machine Learning and Deep Learning Approach." *IEEE PEEIACON*.
- Jeba, S. M., Aurpa, T. T., Siyam, F. H., Khan, R., & Mansia, F. M. (2023). "Analysis of Public Sentiment on Dhaka Metro Rail with Transformer-Based Architectures." *IEEE ICCIT*.
- Google Scholar: scholar.google.com/citations?user=JC4pl4QAAAAJ

CONFERENCE PRESENTATIONS

- Poster Presenter, "BanglishBERT-CNN-MTL," IEEE QPAIN, CUET, Chattogram, Apr 2026.
- Oral Presenter, "Deep Learning Analysis of Public Sentiment on the Padma Bridge Through YouTube Comments," IEEE SPICSCON, University of Rajshahi, Nov 2025.

TECHNICAL SKILLS

- Programming:** Python, R, SQL, C, C++, Java, C#, JavaScript, TypeScript.
- Data Science & ML:** Pandas, NumPy, Scikit-learn, PyTorch, Hugging Face, Matplotlib, exploratory data analysis, statistical analysis, multi-label classification, supervised contrastive learning.
- Software & Data Systems:** ASP.NET Core, Angular 17, SQL Server, REST APIs, Swagger, Git, Jupyter Notebook, Google Colab, Power BI/Tableau.
- Research:** dataset curation, data scraping, experiment design, model evaluation, literature review, scientific writing, poster and oral presentation.

LEADERSHIP & SERVICE

- General Member, IUBAT Innovation and Entrepreneurship Center (IIEC), 2023-2024; supported Techno Carts V.1.
- General Member, Debating Forum of IUBAT, 2023-2024; qualifier, IUBAT Debate Fest 2023.
- General Member, IEEE Student Branch at IUBAT; participated in academic and technical activities.

- Class Representative for multiple semesters; supported student-faculty communication and coordination.
- IUBAT Competitive Programming Community Level 1 Programme, 2023; placed 11th in a selection contest and 19th in a monthly contest.

RESEARCH & GRADUATE STUDY INTERESTS

- Low-resource and multilingual NLP, representation learning, multi-label classification, interpretable machine learning, data-centric AI, responsible and inclusive AI.
- Target intake: Fall 2027 for an M.Sc. in Artificial Intelligence; partial scholarship preferred.
- Long-term goal: build a research-oriented career in AI and Data Science, particularly developing accurate and accessible NLP solutions for low-resource languages such as Bangla.

REFERENCES

- Tanjim Taharat Aurpa, Chairman, Department of Data Science and Engineering, University of Frontier Technology | aurpa0001@bdu.ac.bd | +880 1755-728141
- Md. Saidur Rahman, Assistant Professor, Department of CSE, IUBAT | s.rahman@iubat.edu | +880 1911-729378
- Dr. Md. Alomgir Hossain, Associate Professor, Department of CSE, IUBAT | alomgir.hossain@iubat.edu | +880 1718-735128

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